6IT4-24: Mobile Application Development Lab

Viva Questions & Answers – Complete Guide
Credit: 1.5 | Max. Marks: 100 (IA:60, ETE:40) | End Term Exam: 3 Hours

Experiment 1: Android Studio & Hello World Application

Q1. What is Android?

Ans: Android is an open-source, Linux-based mobile operating system developed by Google. It is used primarily for touchscreen devices like smartphones and tablets.

Q2. What is Android Studio?

Ans: Android Studio is the official Integrated Development Environment (IDE) for Android development, based on IntelliJ IDEA. It provides tools to build, test, and debug Android apps.

Q3. What are the system requirements to install Android Studio?

Ans: Minimum: 8 GB RAM, 8 GB disk space, 1280x800 screen resolution, Windows 8/10/11 64-bit, macOS 10.14+, or Linux. Recommended: 16 GB RAM, SSD storage.

Q4. What is an APK file?

Ans: APK (Android Package Kit) is the file format used by Android to distribute and install apps. It contains the compiled code (.dex), resources, assets, and manifest file.

Q5. What is the Android Manifest file?

Ans: AndroidManifest.xml is a configuration file that contains essential info about the app — package name, activities, permissions, services, and hardware requirements.

Q6. What is the purpose of the R.java file?

Ans: R.java is an auto-generated file by the Android build system. It maps all resources (layouts, strings, drawables) to unique integer IDs used in Java/Kotlin code.

Q7. What is an Activity in Android?

Ans: An Activity represents a single screen in an Android application. It provides a window to interact with the user and has a defined lifecycle.

Q8. What is the difference between SDK and NDK?

Ans: SDK (Software Development Kit) is used to write Android apps in Java/Kotlin. NDK (Native Development Kit) allows writing performance-critical parts in C/C++.

Q9. What is Gradle in Android?

Ans: Gradle is the build system used in Android Studio. It automates compiling, linking, and packaging the Android app into an APK or AAB.

Q10. Name the main folder structure in an Android project.

Ans: app/src/main/java (source code), app/src/main/res (resources), app/src/main/AndroidManifest.xml, Gradle build files (build.gradle, settings.gradle).

Experiment 2: Activity, Intent & Login Module

Q1. What is an Intent in Android?

Ans: An Intent is a messaging object used to request an action from another component. There are two types: Explicit Intent (targets a specific component) and Implicit Intent (declares a general action).

Q2. What is the difference between Explicit and Implicit Intent?

Ans: Explicit Intent specifies the exact component class to start (e.g., startActivity(new Intent(this, SecondActivity.class))). Implicit Intent does not name a specific component, allowing the Android system to find an appropriate component.

Q3. What is the Activity Lifecycle?

Ans: The Activity Lifecycle consists of: onCreate() → onStart() → onResume() → onPause() → onStop() → onDestroy(). Use onRestart() when activity resumes from stopped state.

Q4. What is the use of onCreate() method?

Ans: onCreate() is called when the activity is first created. It is used to initialize components — setContentView(), binding UI elements, and setting up data.

Q5. How do you pass data between activities?

Ans: Using Intent.putExtra(key, value) to send, and getIntent().getStringExtra(key) or getIntExtra(key, default) to receive data in the next activity.

Q6. What is SharedPreferences?

Ans: SharedPreferences is a lightweight key-value storage mechanism in Android to store small amounts of primitive data (booleans, strings, ints) persistently.

Q7. What is the difference between finish() and onBackPressed()?

Ans: finish() programmatically closes the current activity. onBackPressed() is called when the user presses the back button — you can override it to customize behavior.

Q8. What is a Bundle?

Ans: A Bundle is used to pass data between activities and save/restore the instance state of an activity using onSaveInstanceState() and onRestoreInstanceState().

Q9. What is onPause() used for?

Ans: onPause() is called when the activity is partially obscured. It should be used to release resources like camera, sensors, or save unsaved changes that are lightweight.

Q10. What is startActivityForResult()?

Ans: It starts an activity and expects a result back. The receiving activity calls setResult() and the calling activity gets the result in onActivityResult(). (Deprecated in newer APIs, use ActivityResultLauncher instead.)

Experiment 3: GUI Application – Calculator

Q1. What is a Layout in Android?

Ans: A Layout defines the visual structure of a UI. Common layouts: LinearLayout, RelativeLayout, ConstraintLayout, FrameLayout, TableLayout.

Q2. What is the difference between LinearLayout and RelativeLayout?

Ans: LinearLayout arranges views in a single row or column (horizontal/vertical). RelativeLayout positions views relative to each other or the parent container.

Q3. What is ConstraintLayout?

Ans: ConstraintLayout is a flexible layout where you position views using constraints. It is flat (no nested views), improving performance compared to nested LinearLayouts.

Q4. What is the difference between wrap_content and match_parent?

Ans: wrap_content makes the view just big enough to hold its content. match_parent makes it expand to fill the parent container.

Q5. What is an EditText?

Ans: EditText is a UI widget that allows users to enter and edit text. It extends TextView and adds editable functionality.

Q6. What is the difference between TextView and EditText?

Ans: TextView only displays text (read-only). EditText is an editable version that allows user input.

Q7. How do you handle button click events?

Ans: Using setOnClickListener() or by defining the onClick attribute in XML. Example: button.setOnClickListener(v -> { /* action */ });

Q8. What is Toast in Android?

Ans: Toast is a small popup message shown briefly to the user. Example: Toast.makeText(context, 'Message', Toast.LENGTH_SHORT).show();

Q9. What is a Spinner?

Ans: Spinner is a dropdown menu that allows users to select one option from a list. It is similar to a combo box in desktop applications.

Q10. What is dp vs sp vs px in Android?

Ans: dp (density-independent pixels) is used for layout dimensions. sp (scale-independent pixels) is used for font sizes. px is actual screen pixels — not recommended for use directly.

Experiment 4: RSS Feed Application

Q1. What is RSS Feed?

Ans: RSS (Really Simple Syndication) is a web feed format used to publish frequently updated content. It uses XML format and allows users to receive updates from multiple sources.

Q2. What is XML Parsing in Android?

Ans: Android supports SAX, DOM, and XmlPullParser for XML parsing. XmlPullParser is recommended as it is efficient and event-driven.

Q3. What is AsyncTask?

Ans: AsyncTask (deprecated in API 30) is used to perform background operations and publish results on the UI thread. It has doInBackground(), onPreExecute(), and onPostExecute() methods.

Q4. What replaced AsyncTask in modern Android?

Ans: AsyncTask is replaced by Kotlin Coroutines, ExecutorService with Handler, or WorkManager for background tasks.

Q5. What is a RecyclerView?

Ans: RecyclerView is an advanced widget to display large sets of data efficiently. It recycles item views using ViewHolder pattern, improving memory performance compared to ListView.

Q6. What is the role of Adapter in Android?

Ans: Adapter acts as a bridge between data source and UI (ListView, RecyclerView). It creates view items and binds data to them.

Q7. What permissions are needed for internet access in Android?

Ans: You must declare in AndroidManifest.xml.

Q8. What is Volley library?

Ans: Volley is an HTTP library by Google for Android that simplifies network requests — supports GET, POST, JSON, image requests with built-in caching.

Q9. What is the difference between ListView and RecyclerView?

Ans: ListView is simpler but less efficient. RecyclerView provides better performance using ViewHolder, supports animations, multiple layout managers, and item decorators.

Q10. What is JSON?

Ans: JSON (JavaScript Object Notation) is a lightweight data interchange format. In Android, use JSONObject and JSONArray to parse JSON data, or libraries like Gson/Moshi.

Experiment 5: Drawing Graphical Primitives

Q1. What is a Canvas in Android?

Ans: Canvas is a 2D drawing surface. You can draw shapes, text, images, and paths on it. It provides methods like drawCircle(), drawRect(), drawLine(), drawText().

Q2. What is a Paint object?

Ans: Paint holds style and color information about how to draw geometries, text, and bitmaps. You can set color, stroke width, text size, antiAlias, etc.

Q3. How do you create a custom View?

Ans: Extend the View class, override onDraw(Canvas canvas) to do custom drawing, and override onMeasure() if needed. Then add it to a layout.

Q4. What is the difference between View and SurfaceView?

Ans: View is drawn on the UI thread and suitable for static/simple animations. SurfaceView has a dedicated drawing thread, suitable for games and real-time graphics.

Q5. What is onDraw() method?

Ans: onDraw(Canvas canvas) is called by the system to draw the view. Override it to perform custom drawing using Canvas and Paint objects.

Q6. What are different shapes you can draw on Canvas?

Ans: Canvas supports: drawRect(), drawCircle(), drawOval(), drawLine(), drawArc(), drawPath(), drawBitmap(), drawText(), drawRoundRect().

Q7. What is Bitmap in Android?

Ans: Bitmap represents a mutable or immutable image. It stores pixel data and can be drawn on a Canvas. BitmapFactory is used to decode images from files/resources.

Q8. What is Path in Android drawing?

Ans: Path is used to define complex geometric shapes by combining lines, curves, and arcs. Draw it using canvas.drawPath(path, paint).

Q9. What is invalidate() in Android?

Ans: invalidate() is called to request the system to redraw the view by calling onDraw() again. It must be called on the UI thread. Use postInvalidate() from background threads.

Q10. What is anti-aliasing in Android graphics?

Ans: Anti-aliasing smooths the edges of shapes and text by blending colors at the boundary. Enable it with paint.setAntiAlias(true).

Experiment 6: SQLite Database

Q1. What is SQLite?

Ans: SQLite is a lightweight, embedded relational database management system (RDBMS) used in Android. It stores data in a local .db file on the device's internal storage.

Q2. What is SQLiteOpenHelper?

Ans: SQLiteOpenHelper is an abstract class that helps create, upgrade, and manage database versions. Override onCreate() to create tables and onUpgrade() to handle version changes.

Q3. What are the four basic CRUD operations in SQLite?

Ans: CREATE (INSERT), READ (SELECT/query), UPDATE (UPDATE), DELETE (DELETE). These are performed using SQLiteDatabase methods or raw SQL queries.

Q4. What is ContentValues?

Ans: ContentValues is a key-value map used to insert or update data in SQLite. Keys are column names and values are the data to insert.

Q5. What is the difference between execSQL() and rawQuery()?

Ans: execSQL() executes SQL statements that don't return data (INSERT, UPDATE, DELETE, CREATE). rawQuery() executes SELECT statements and returns a Cursor.

Q6. What is a Cursor in Android?

Ans: Cursor is an interface that provides random read-write access to the result set returned by a database query. Use moveToFirst(), moveToNext(), getString(), getInt() to navigate and read data.

Q7. What is the db.query() method?

Ans: db.query(table, columns, selection, selectionArgs, groupBy, having, orderBy) is a structured way to run SELECT queries without writing raw SQL.

Q8. How do you upgrade a database schema?

Ans: Increase the database version number. Override onUpgrade() in SQLiteOpenHelper to ALTER TABLE or DROP and recreate tables, then migrate data.

Q9. What is Room Database?

Ans: Room is an Android Jetpack library that provides an abstraction layer over SQLite. It uses annotations (@Entity, @Dao, @Database) to simplify database operations with compile-time checks.

Q10. Where is the SQLite database stored on an Android device?

Ans: The database is stored in /data/data//databases/ directory on the device's internal storage.

Experiment 7: GPS Location Application

Q1. What is GPS?

Ans: GPS (Global Positioning System) is a satellite-based navigation system that provides location (latitude, longitude) and time information anywhere on Earth.

Q2. What permissions are required for GPS in Android?

Ans: ACCESS_FINE_LOCATION (GPS/Wi-Fi precise location) and ACCESS_COARSE_LOCATION (network-based approximate location) in AndroidManifest.xml. Runtime permission needed from Android 6.0+.

Q3. What is LocationManager?

Ans: LocationManager is an Android system service that provides access to location services. Use it to request location updates via GPS_PROVIDER or NETWORK_PROVIDER.

Q4. What is the difference between GPS_PROVIDER and NETWORK_PROVIDER?

Ans: GPS_PROVIDER uses satellite signals — more accurate but slower and battery-heavy. NETWORK_PROVIDER uses Wi-Fi/cell towers — faster and less battery consumption but less accurate.

Q5. What is LocationListener?

Ans: LocationListener is an interface with callbacks: onLocationChanged(), onStatusChanged(), onProviderEnabled(), onProviderDisabled() — used to receive location updates.

Q6. What is FusedLocationProviderClient?

Ans: FusedLocationProviderClient (Google Play Services) intelligently combines GPS, Wi-Fi, and cell network to provide accurate location efficiently with less battery usage.

Q7. What is Geocoder?

Ans: Geocoder converts coordinates (lat, lng) to human-readable addresses (reverse geocoding) and vice versa (forward geocoding).

Q8. What is requestLocationUpdates()?

Ans: locationManager.requestLocationUpdates(provider, minTime, minDistance, listener) starts delivering location updates to the listener based on time/distance thresholds.

Q9. How do you display location on a Map?

Ans: Use Google Maps Android API. Obtain API key from Google Cloud Console, add the Maps SDK dependency, use SupportMapFragment, and place a Marker at the obtained LatLng.

Q10. What is a Geofence?

Ans: A Geofence is a virtual geographic boundary. Android Geofencing API triggers alerts when a device enters, exits, or dwells within a defined circular area.

Experiment 8: Writing Data to SD Card

Q1. What is external storage in Android?

Ans: External storage is a shared storage space (SD card or emulated). Files stored here are accessible to other apps and the user via a file manager.

Q2. What permissions are needed to read/write SD card?

Ans: READ_EXTERNAL_STORAGE and WRITE_EXTERNAL_STORAGE in Manifest. Runtime permission required from Android 6.0+. From Android 10+, Scoped Storage rules apply.

Q3. What is Scoped Storage?

Ans: Scoped Storage (Android 10+) restricts apps from freely accessing external storage. Apps can only access their own app-specific directory (getExternalFilesDir()) without permissions.

Q4. What is the difference between internal and external storage?

Ans: Internal storage is private to the app, not accessible by others. External storage (SD card/shared) is accessible to all apps and the user but may be removed.

Q5. What is FileOutputStream?

Ans: FileOutputStream writes bytes to a file. Used with openFileOutput() for internal storage or new FileOutputStream(file) for external storage.

Q6. What is FileInputStream?

Ans: FileInputStream reads bytes from a file. Use openFileInput() for internal storage files or new FileInputStream(file) for external.

Q7. How do you check if external storage is available?

Ans: Use Environment.getExternalStorageState(). Check if it equals Environment.MEDIA_MOUNTED for read/write access.

Q8. What is Environment.getExternalStorageDirectory()?

Ans: It returns the primary external storage directory. Deprecated in API 29 — use Context.getExternalFilesDir() instead for scoped access.

Q9. What is a FileProvider?

Ans: FileProvider is a ContentProvider that securely shares files with other apps using content:// URIs instead of file:// URIs, required from Android 7.0+.

Q10. What is the difference between getFilesDir() and getExternalFilesDir()?

Ans: getFilesDir() returns app-specific internal storage directory. getExternalFilesDir() returns app-specific external storage directory — both are private but external can be accessed if SD card is mounted.

Experiment 9: Gaming Application

Q1. What is a Game Loop?

Ans: A Game Loop is a continuous loop that updates the game state (physics, logic) and renders frames. In Android, implemented using SurfaceView with a background thread.

Q2. What is SurfaceView used for in gaming?

Ans: SurfaceView provides a dedicated drawing surface in a separate thread, allowing rendering without blocking the main UI thread — ideal for games.

Q3. What is SurfaceHolder?

Ans: SurfaceHolder is an interface to control the surface. Use surfaceHolder.lockCanvas() to get Canvas for drawing and unlockCanvasAndPost() to render the frame.

Q4. What is the difference between SurfaceView and TextureView?

Ans: SurfaceView has its own surface in the Z-order (below/above window). TextureView is part of the view hierarchy and supports transformations like rotation/scaling — better for video.

Q5. What are Sprites in gaming?

Ans: Sprites are 2D images/animations used to represent game objects (player, enemies, bullets). In Android, managed as Bitmaps drawn on Canvas at specific positions.

Q6. What is collision detection?

Ans: Collision detection checks whether two game objects overlap or touch. Simple method: use Rect.intersects() to detect rectangular bounding box collisions.

Q7. How do you handle touch events in a game?

Ans: Override onTouchEvent(MotionEvent event) in SurfaceView. MotionEvent provides action (ACTION_DOWN, ACTION_MOVE, ACTION_UP) and coordinates.

Q8. What is MediaPlayer in Android?

Ans: MediaPlayer is used to play audio and video files. For games, use it to play background music or SoundPool for short sound effects with low latency.

Q9. What is SoundPool?

Ans: SoundPool is used to load and play short audio clips simultaneously with low latency — ideal for game sound effects like explosions, jumps.

Q10. What is a Handler in Android?

Ans: Handler is used for communication between background threads and the main UI thread. In games, post UI updates (score display) using handler.post() or handler.sendMessage().

Experiment 10: Image and Video Handling

Q1. What is Glide library?

Ans: Glide is a fast and efficient image loading and caching library for Android. It handles image loading from URLs, files, or resources with automatic caching and memory management.

Q2. What is Picasso?

Ans: Picasso is another popular image loading library by Square. It simplifies image loading into ImageView with automatic memory/disk caching.

Q3. How do you capture an image using Camera in Android?

Ans: Use an Intent with MediaStore.ACTION_IMAGE_CAPTURE to open the default camera app. The captured image is returned in onActivityResult() as a Bitmap or saved to a URI.

Q4. What is MediaStore?

Ans: MediaStore is Android's content provider that organizes media files (images, videos, audio). Used to query and insert media files into the device's media library.

Q5. What is VideoView?

Ans: VideoView is a UI component to play video files. It supports local files and streaming URLs. Use MediaController to add playback controls.

Q6. What is ExoPlayer?

Ans: ExoPlayer is Google's advanced open-source media player library. It supports DASH, HLS streaming, DRM, and more — recommended over MediaPlayer for complex media needs.

Q7. How do you resize an image in Android?

Ans: Use Bitmap.createScaledBitmap(original, width, height, filter) to scale a bitmap. Or use BitmapFactory.Options.inSampleSize to downsample large images on load.

Q8. What is BitmapFactory?

Ans: BitmapFactory creates Bitmap objects from files, byte arrays, or resources. Use BitmapFactory.Options to control decoding, such as inSampleSize for memory efficiency.

Q9. What is a URI in Android?

Ans: URI (Uniform Resource Identifier) identifies a resource. In Android, content:// URIs are used to reference files in ContentProviders (MediaStore). file:// URIs reference file system paths.

Q10. What is Scoped Storage impact on media access?

Ans: From Android 10+, apps must use MediaStore API or Storage Access Framework to access shared media. Direct file path access is restricted. Apps must request READ_MEDIA_IMAGES/VIDEO permissions from Android 13+.

★ MOST IMPORTANT VIVA QUESTIONS (All Experiments)

★ Q1. What is the difference between Service and Activity?

Ans: Activity has a UI and represents a screen. Service runs in the background without a UI to perform long-running operations like downloading files, playing music.

★ Q2. What is a Fragment?

Ans: A Fragment is a reusable portion of a UI that lives inside an Activity. It has its own lifecycle and can be combined in multi-pane layouts. Example: navigation tabs.

★ Q3. What is the difference between Serializable and Parcelable?

Ans: Both pass objects between Activities. Parcelable is Android-specific, faster, and more efficient. Serializable is Java standard, slower due to reflection — avoid for Android.

★ Q4. What is ANR in Android?

Ans: ANR (Application Not Responding) occurs when the main UI thread is blocked for more than 5 seconds (broadcast: 10 seconds). Always perform heavy operations in background threads.

★ Q5. What is Context in Android?

Ans: Context is an abstract class that provides access to app resources, system services, and components. Types: Application Context (app lifecycle) and Activity Context (activity lifecycle).

★ Q6. What is a BroadcastReceiver?

Ans: BroadcastReceiver listens for system-wide or app-specific broadcast messages (battery low, SMS received, network change). Register in Manifest (static) or in code (dynamic).

★ Q7. What is the difference between startService() and bindService()?

Ans: startService() starts an independent service that runs until stopped. bindService() binds to a service for interaction — service runs as long as a component is bound.

★ Q8. What is a ContentProvider?

Ans: ContentProvider manages access to a structured set of data, allowing data sharing between apps. Accessed using ContentResolver. Used in MediaStore, Contacts.

★ Q9. What is WorkManager?

Ans: WorkManager is a Jetpack library for guaranteed background work. It replaces JobScheduler, AlarmManager for deferrable, guaranteed tasks — respects Doze mode and battery restrictions.

★ Q10. What is the difference between compileSdkVersion, minSdkVersion, and targetSdkVersion?

Ans: compileSdkVersion: Android API version used to compile. minSdkVersion: lowest Android version the app supports. targetSdkVersion: API version the app is designed and tested for.

★ Q11. What is ViewModel in Android?

Ans: ViewModel is a Jetpack component that stores and manages UI-related data. It survives configuration changes (rotation) unlike Activity, preventing data loss.

★ Q12. What is LiveData?

Ans: LiveData is a lifecycle-aware observable data holder. It notifies observers (UI) of data changes only when they are in active state — prevents memory leaks and crashes.

★ Q13. What is the difference between HTTP GET and POST?

Ans: GET requests data from a server (parameters in URL, visible, limited size). POST sends data to server (in request body, not visible in URL, no size limit).

★ Q14. What is Retrofit?

Ans: Retrofit is a type-safe HTTP client library for Android by Square. It simplifies API calls by converting HTTP API into Java/Kotlin interfaces with annotations.

★ Q15. What is the purpose of build.gradle in Android?

Ans: build.gradle defines the build configuration: dependencies, SDK versions, build types (debug/release), flavors, signing configs, and plugins for the Android project.